# 42088 – Industrial Project

# Project Status Report

## Milestone 3 – Construction I

| | |
|---|---|
| Project name: | Automatic, universal push-button actuator, equipped with force gauge |
| Customer: | Altice Labs |
| Customer supervisor: | Eng. Hélder Alves<br>Eng. Nuno Balseiro |
| Date issued: | 27/11/2024 |
| Team members: | Coordinator: Cláudia Almeida, claudiasalmeida@ua.pt, 914102617.<br>Other team members:<br>Rafael Alves, rafael01@ua.pt , 937353627.<br>Gonçalo Peralta, goncaloperalta@ua.pt , 916828627.<br>Pedro Matias, pedrodsmatias@ua.pt , 917126548. |
| Supervisor: | Professor Pedro Fonseca |
| Class supervisor: | Professor Pedro Fonseca |

## Management summary:

The project is on track to meet its delivery deadlines. The team has successfully completed the elaboration phase and is now transitioning into the construction phase of the project. At this stage, all of the necessary hardware components have been delivered. The PCBs required to power and control the actuator have been thoroughly designed and reviewed multiple times with guidance and feedback from both class and customer supervisor. Additionally, the first draft of the user interface was presented using simulated data to demonstrate functionality. Feedback was collected during this phase, and improvements are being incorporated to refine the interface for its next iteration.

As we move forward, our focus will shift to establishing communication with the Raspberry Pi, generating test data, processing feedback signals from the test gateway and start individual tests on main components such as the actuator and the force sensor. Meanwhile, we await the printing of the PCBs, after which we will proceed with assembling and soldering all the components. This marks an exciting and critical step in bringing the system closer to full functionality.

Considering all the progress and development to date, as well as the organization of future tasks, the team anticipates that all proposed objectives will be successfully achieved.

# Architecture

The architecture of the system consists of three main components: the **Control Unit**, the **Control Electric Circuit**, and the **Mechanical Actuation**.

- The **Control unit** is managed by a Raspberry Pi that runs the control software and serves as the central processing component for the system. The Raspberry Pi generates a PWM (Pulse Width Modulation) signal to regulate the actuator's movement, allowing for controlled button pressing with variable force. It also maintains a Secure Shell (SSH) connection with the DUT, allowing it to detect feedback from the device in real time, such as system responses or status changes upon a button click. For force measurement, the Raspberry Pi reads data from the force sensor using I2C/SPI protocols, ensuring accurate and timely data transfer from the sensor to the control software.
- The **Control electric circuit** amplifies the PWM signal generated by the Raspberry Pi, ensuring sufficient power is supplied to the actuator. This circuit includes an **H-Bridge**, allowing current to flow in both directions as required to control the actuator's movement accurately. The circuit, powered by a 12V, 3A DC power source, will provide the energy needed to drive the actuator with stability and precision, allowing for both pressing and releasing movements, required for testing various types of buttons.
- The **Mechanical Actuation** assembly is designed to interface directly with the DUT, positioning the actuator over the buttons to be tested. This component includes a clamp that securely attaches to the DUT, providing flexibility and adaptability for different button placements and device sizes. Mounted within the clamp is a linear actuator that performs the physical pressing of the buttons, with a force sensor placed behind the actuator to capture the exact force applied during each press. This assembly connects to a PCB board, which relays signals from the force sensor back to the Control Unit, ensuring accurate control and data capture throughout the testing process.

Figure 1 shows the improved and updated mechanical design of the product.

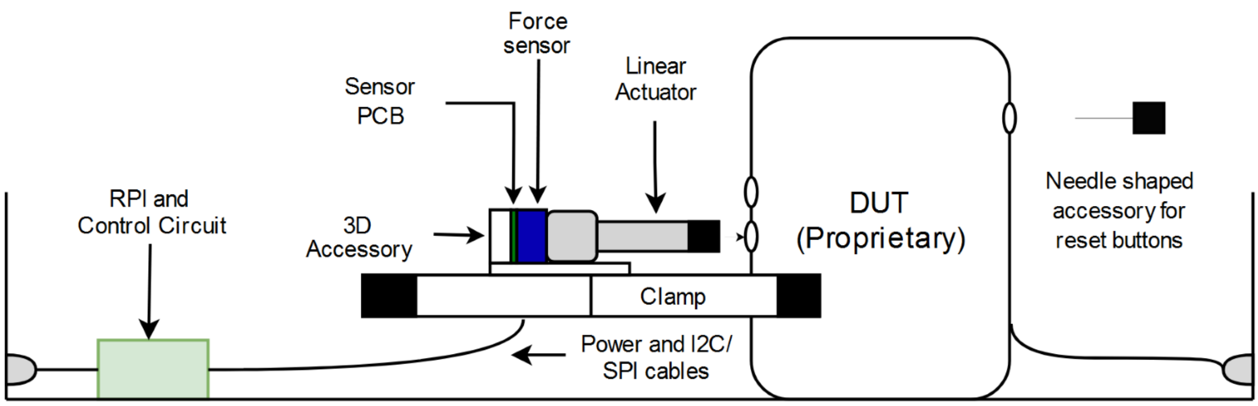
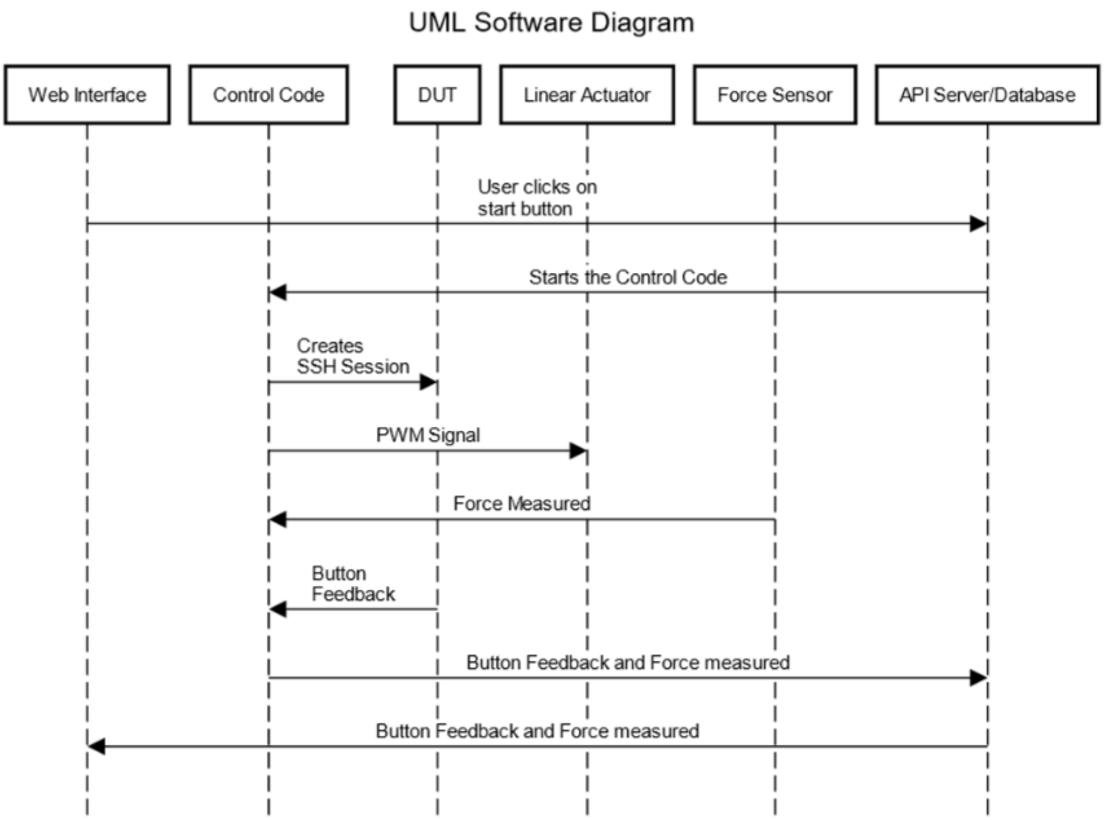



**Figure 1.** Mechanical assembly of the product.

Figure 2 shows the Unified Modelling Language (UML) Sequence Diagram, which represents the interaction amongst the different components of the system.

**Figure 2.** UML Diagram.




Further details on the system's architecture can be found in the attached document "BlockDiagrams".

# Key milestones

| ID | Description | Original planned completion date | Current planned completion date | Actual completion date |
|---|---|---|---|---|
| **Software Web Interface** | Suggested visual presentation of the web interface | 13/11/2024 | 13/11/2024 | 14/11/2024 |
| **Hardware PCB's design** | Design of the system's printed circuits and electrical circuits | 13/11/2024 | 18/11/2024 | 21/11/2024 |
| **Project Status** | A summary of the current | 27/11/2024 | 27/11/2024 | 27/11/2024 |

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| **Communication with Raspberry Pi** | Generate test signals and process feedback signals from the DUT. | 04/12/2024 | 04/12/2024 | 27/11/2024 |
| **Design of the mechanical attachment** | Development of the mechanical couple using 3D print. | 04/12/2024 | 04/12/2024 | --/--/-- |
| **Individual tests to the main components** | Functionality tests for the main components of the project: force sensor and linear actuator. | 04/12/2024 | 04/12/2024 | --/--/-- |

## Progress and deviation from plan

The development of the PCBs required extra effort to reach a level deemed suitable for the prototyping stage. This additional effort caused a small delay in finalizing the design, leading to a late printing order and, consequently, a significant portion of the testing will be delayed, temporarily stalling progress on the hardware side. To mitigate this deviation, the team will redirect their time to other tasks in two primary fronts: the first one being establishing communication with the Raspberry Pi, generating test data, and processing feedback signals from the test gateway, and the second one consisting on the designing of the mechanical attachment, to be developed using 3D printing.

## Work plan for next iteration

For the next iteration, individual tests on the actuator will be carried out. Additionally, the team will begin analysing and looking into the gateway's feedback signals through given credentials to access its operating system.

Table 2 shows the sprint plan updated.

| Sprints | Weeks | Tasks | | Claúdia | Rafael | Pedro | Gonçalo |
|---|---|---|---|---|---|---|---|
| 1 | 4 | Detailed system architecture and list of components finalized | | x | x | x | x |
| | | PCB design for the control signal and the Force Sensor | Web page development and structurization of a control code on the Rpi | x | x | x | x |
| MVP's | | Web page interface and all PCB's | | We are currently here | | | |
| 2 | 3 | Prototype a 3D support | Test basic Actuator control and SSH connection to the gateway with the RPi | x | x | x | x |
| | | Test the force sensor PCB | Test the controller PCB | x | x | x | x |
| | | Merge everthing into a single control code | | x | x | x | x |
| 3 | 3 | Develop a 3D model to support the architecture | Update the interface to match the control code | x | x | x | x |
| | | Test the overall architecture | | x | x | x | x |
| 4 | 2 | Collecting final feedback before final construction | | x | x | x | x |
| | | Delivery of the prototype to the client | | x | x | x | x |

**Figure 3.** Sprint plan

# Financial status

With respect to the project budget and components list, no modifications have been made since the previous milestone. Consequently, the order was placed as planned, and all required components have now been successfully delivered.

| Quantity | Description | | Status | Price Per Unite(€) | Total Value(€) | % |
|---|---|---|---|---|---|---|
| 1 | Honeywell FMAMSDXX015WC2C3 | Force Sensor | Purchased | 33,67 | 33,67 | 18% |
| 1 | Actuonix L12-50-50-12-S | Actuator | Purchased | 83,66 | 83,66 | 45% |
| 2 | JST Comercial RE-H022TD-1190 | Actuator connector to PCB | Purchased | 0,18 | 0,36 | 0% |
| 1 | CUI Inc. SMM30-12-K-P6 | Power Supplier | Purchased | 17,76 | 17,76 | 10% |
| 1 | Same Sky PJ-079BH | Connector to Power Supplier | Purchased | 1,5 | 1,5 | 1% |
| 1 | Raspberry Pi 4 B | Processing Unit | Purchased | 32,55 | 32,55 | 17% |
| 1 | ST L6205D | Controler for the actuator | Purchased | 6,12 | 6,12 | 3% |
| 1 | Nichicon GYF1H101MCQ1GS | Capacitor | Purchased | 1,11 | 1,11 | 1% |
| 1 | KYOCERA AVX KAM21BR71H104JT | Capacitor | Purchased | 0,19 | 0,19 | 0% |
| 1 | TAIYO YUDEN MAASU21GSB7224KTCA01 | Bootstrap Capacitor | Purchased | 0,4 | 0,4 | 0% |
| 1 | KYOCERA AVX KGF15AR71H103KT | Bootstrap Capacitor | Purchased | 0,23 | 0,23 | 0% |
| 2 | KYOCERA AVX 12065A562FAT2A | Capacitor | Purchased | 0,94 | 1,88 | 1% |
| 2 | onsemi / Fairchild 1N4148 | Bootstrap Diode | Purchased | 0,09 | 0,18 | 0% |
| 2 | Panasonic ERA-8VRB1003V | Resistor | Purchased | 1,18 | 2,36 | 1% |
| 1 | Vishay CRCW2010100RJNEFIF | Bootstrap Resistor | Purchased | 0,55 | 0,55 | 0% |
| 1 | TI INA381A3 | Current Amplifier | Purchased | 1,1 | 1,1 | 1% |
| 1 | TE Connectivity RLC73K2ER162FTDF | Current Measuring Resistor | Purchased | 0,46 | 0,46 | 0% |
| 1 | Vishay MCS0402MD1002BE100 | Pull-Up Resistor | Purchased | 0,45 | 0,45 | 0% |
| 1 | CTS Electronic Components 14TR08SA103CP | Potentiometer | Purchased | 0,71 | 0,71 | 0% |
| 1 | TI TLV9032QDGKRQ1 | Capacitor | Purchased | 1,5 | 1,5 | 1% |
| 1 | KYOCERA AVX KAM21BR71H104JT | Capacitor | Purchased | 0,19 | 0,19 | 0% |
| | | | | Total: | 186,93 | 100% |

**Figure 4.** Updated project budget and final components table.

# Group contribution

| Group member | Major contributions | Work share (%) |
|---|---|---|
| Cláudia Almeida | Team coordinator, responsible for organizing regrouping sessions and managing communication with external parties. Primary editor of written documents, developed the use cases, studied market relevance for industrial automation, and led Milestone 1 presentation. Web interface Development. | 25 |
| Rafael Alves | Developed the final hardware components list, analysed project risks for each phase and proposed mitigation strategies, and led Milestone 1 presentation. Development of the PCB. | 25 |
| Gonçalo Peralta | Created all system diagrams and drafts of the mechanical product for documentation, developed the block diagrams and use cases, sprints spread calendar and led the initial planning for the software approach to the project. Web interface Development. | 25 |
| Pedro Matias | Contributed to researching and finalizing the hardware components list and supported the assessment of technical needs for project phases. Development of the PCB. | 25 |

# Comments and remarks

Attached are the documents relating to the risk analysis of the project and the block diagrams of the system broken down into various levels.